

Forensics Guide: Wireshark Analysis for Network Forensics

What is Network Forensics?

Network Forensics is the process of capturing, recording, and analyzing network traffic to investigate security incidents, data breaches, malware infections, or policy violations.

Unlike single-protocol challenges, forensics scenarios usually involve **multiple protocols** working together in one capture file. You must correlate evidence across layers to reconstruct what happened.

Why It Matters

- Real-world investigations rarely involve just one protocol.
- You learn to think like a Security Analyst / Incident Responder.
- This category ties together everything you learned from HTTP, TCP, DNS, FTP, ICMP, SMTP, and TLS.

Core Analysis Mindset

The Investigative Process:

1. **Understand the big picture** — What protocols are present?
2. **Identify key events** — Logins, file transfers, connections, DNS queries.
3. **Build a timeline** — What happened and in what order?
4. **Correlate evidence** — Link DNS → TCP → HTTP → FTP, etc.
5. **Find the flag** — Usually well-hidden across multiple clues.

Essential Wireshark Techniques for Forensics

1. Initial Reconnaissance Filters

Bash

```
dns || http || ftp || smtp || icmp || tls  
tcp.flags.syn == 1      # New connections  
http.request or ftp.request or smtp.command
```

2. Statistics & Overview Tools

- **Statistics** → **Protocol Hierarchy** — See which protocols are present and their volume.
- **Statistics** → **Conversations** (TCP, UDP, IPv4) — Top talkers.
- **Statistics** → **Endpoints** — All IP addresses involved.
- **Statistics** → **DNS / HTTP / TLS** — Protocol-specific summaries.
- **IO Graph** — Traffic patterns over time.

3. Timeline Building

- Sort by **Time** column.
- Use **Time Reference** (right-click → Set Time Reference) to mark key events.
- Colorize rules to highlight important packets.

4. Advanced Search Techniques

- frame contains "THAGHRAH" or frame contains "flag"
- Search across multiple protocols at once.
- Use **Expert Information** (yellow/red warnings).

Common Network Forensics Scenarios

1. **Data Exfiltration**
 - DNS tunneling + large ICMP packets
 - FTP/HTTP upload of sensitive files
 - Email attachments (SMTP)
2. **Credential Theft**
 - Plaintext logins (FTP, SMTP, HTTP)
 - Session cookie theft via HTTP
3. **Malware Beaconing**
 - Periodic DNS or HTTP requests
 - Suspicious User-Agent strings
4. **Reconnaissance**
 - Port scanning (many SYN packets)
 - ICMP sweeps + DNS queries
5. **Lateral Movement**
 - Multiple internal connections after initial breach
6. **Covert Channels**
 - Data hidden in ICMP, DNS TXT, or unusual TLS extensions

Practical Forensics Workflow

1. **Open PCAP** → Apply broad filter and study **Protocol Hierarchy**.
2. **Identify all hosts** involved using Conversations/Endpoints.
3. **Follow important TCP streams** (especially long ones).
4. **Extract objects**:
 - File → Export Objects → HTTP
 - File → Export Objects → FTP-DATA
 - File → Export Objects → SMB (if present)
5. **Reconstruct user actions** step by step.
6. **Hunt for the flag** using broad text searches.
7. **Document findings** — timestamps, IPs, protocols, and artifacts.

Example Challenge Goals in Forensics Category

- Reconstruct an entire attack chain (recon → breach → data theft).

- Find stolen credentials used across multiple protocols.
- Recover a deleted or transferred sensitive file.
- Identify the attacker's Command & Control (C2) server.
- Locate a flag split across DNS, HTTP, and ICMP packets.
- Determine how data was exfiltrated from the network.

Pro Tips

- Use **** coloring rules**** to quickly spot suspicious activity.
- Export suspicious files and analyze them outside Wireshark.
- Pay attention to **timestamps** — small time gaps can be significant.
- When stuck, search for the flag string broadly (Edit → Find Packet).
- Think about **what doesn't belong** — unusual traffic is often the key.